UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Data representation and number systems

Calculate -86 - (-68) using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

Answer: left -86 -> 1 1010110; right -68 -> 1 1000100; -86 - (-68) = -18 -> sign-magnitude 1 0010010

- *1 mark for converting both operands to sign-magnitude.*
- *1 mark for applying the sign rule for subtraction.*
- *1 mark for the correct magnitude operation.*
- *1 mark for the correct sign bit.*
- *1 mark for the final decimal and byte representation.*

QUESTION A2 (5 marks)

Topic: Parallel and multiprocessor architecture

a) List TWO (2) advantages of SMP organization.
b) Define cluster organization.
c) Give ONE (1) benefit of cluster organization.

Source: CSC429 (7).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Computer generations and history

The advancement of technology through different computer generations has focused on improving performance, speed, and efficiency. From vacuum tubes in the first generation to modern AI-powered systems, each generation introduced faster processing, smaller components, and greater reliability.

a) Name the inventor of the following devices:
i)
Mechanical Calculating Machine
ii)
Analytical Machine
iiD
IAS Machine
b) Why are the machines of later computer generation designed for general purposes instead of the eadier computer generation that is more specific in function?

Source: CSC429 (20).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)

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List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Computer architecture and CPU organisation

a) Compare system interconnection and CPU interconnection in terms of:
i) function
ii) physical appearance.
b) Describe the function of the THREE (3) different bus types in a computer system.

Source: CSC429 (16).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Memory hierarchy and cache

a) List FOUR (4) types of access method for storage device with ONE (1) example for each of them.
b) Calculate the total of memory location using 11 address bit.
c) Calculate the address bits required for 2MB of RAM.

Source: CSC429 (10).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Input/output and interrupts

a)
Define interrupt.
b)
Differentiate between interrupt and direct manipulation access.
c)
Given the following scenario.
Dilan is surfing woolworth.com to buy groceries online. While he is comparing prices of products in woolworth.com and one other website, two messages popped up onto his screen. Figure 2 (a) and (b) illustrate the conveyed messages.

Are you sure you want to
cancel this request?

(a)

There is a hardware event failure
in this computer. Please reboot
this computer now.

(b)

Figure 2: The pop-up messages

i)
Identify the type of interrupt in Figure 2(a) and Figure 2(b) with proper justification.
ii)
While interruptroutinefor Interrupt Ais executing, it is interrupted by interrupt Bwhich has a higher priority than interrupt A. What will happen to interrupt routine A?

Source: CSC429 (13).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

Keyboard is the most common means for computer interaction.

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a) Explain TWO (2) types of keyboard characters.
b) Describe THREE (3) categories of input and output module. Give an appropriate example for each category.

Source: CSC429 (2).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

a) Differentiate between organization and architecture.
b) Explain the function of THREE (3) Central Processing Unit (CPU) components.

Source: CSC429 (6).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: 50 - (-64)

a) Calculate 50 - (-64) using sign and magnitude.

Answer: left 50 -> 0 0110010; right -64 -> 1 1000000; 50 - (-64) = 114 -> sign-magnitude 0 1110010

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate 50 - (-64) using 1's and 2's complement.

Answer: left 50 -> 1's 00110010, 2's 00110010; right -64 -> 1's 10111111, 2's 11000000; 50 - (-64) = 114 -> 1's 01110010, 2's 01110010

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.

y = A -
2C
* D
B + A2

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

Three-address	One-address	Two-address
LOADp		
MOV (p,q)		
ADD (p.q.r)		
STORp		
ADD (p,q)		
SUB (p,q,r)		
ADDp		
SUB (p,q)		
MUL (p.q.r)		
SUBp		
MUL (p,q)		
DIV(p,q,r)		
MULp		
DIV(p,q)		
DIVp		

Source: CSC429 (18).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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